Artificial Intelligence Project - Pacman Game

Project Member

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Project Idea

Designing a simplified Pacman game using Python

Project Description

Pac-Man is an intelligent game agent that navigates through a maze to collect food pellets while avoiding ghosts.
The project applies Artificial Intelligence techniques such as pathfinding, search algorithms, and predictive ghost avoidance to enable autonomous decision making.

Agent Architecture

- **Model-Based Reflex Agent**
- **Goal-Based Agent**

algorithms used

- A* Search Algorithm
- Breadth-First Search

PEAS Model

Performance Measures

- Maximize collected food pellets
- Avoid ghost collisions
- Complete the maze successfully
- Maximize score

Environment

- **Maze grid**
- **Ghosts**
- **Walls**
- **Food pellets**

Actuators

- **Move Up**
- **Move Down**
- **Move Left**
- **Move Right**

Sensors

- **Pac-Man current position**
- **Ghost positions**
- **Food locations**
- **Danger prediction map**
- **Distance from ghosts**

ODESA Analysis

Observability

- **Fully Observable.**

Determinism

- **Stochastic**

Episodic/Sequential

- **Sequential.**

Environment Type

- **Dynamic**

Agent Type.

- **Multi-Agent**

State Space

- **Discrete**